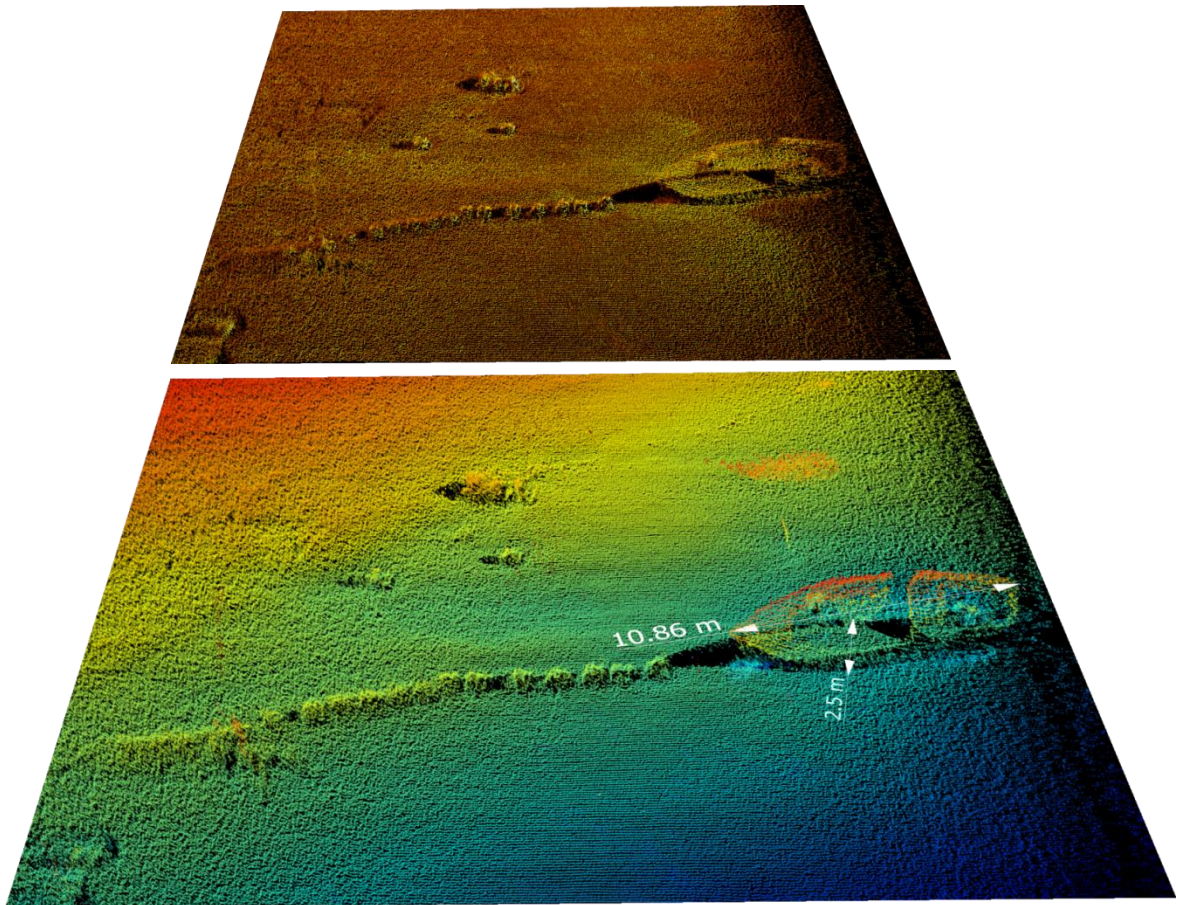
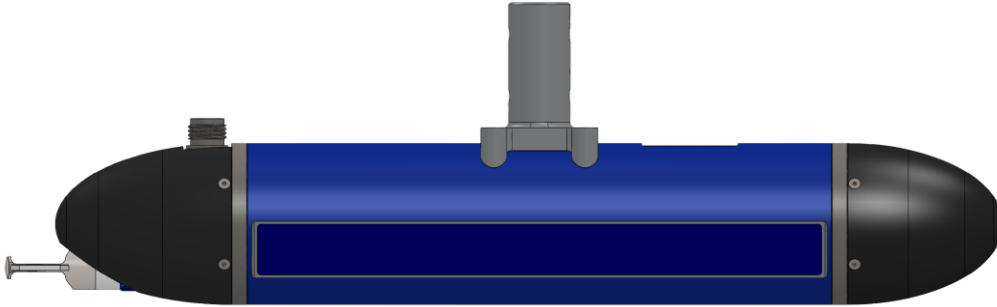


3DSS-IDX Quick Start Guide





1 SUMMARY

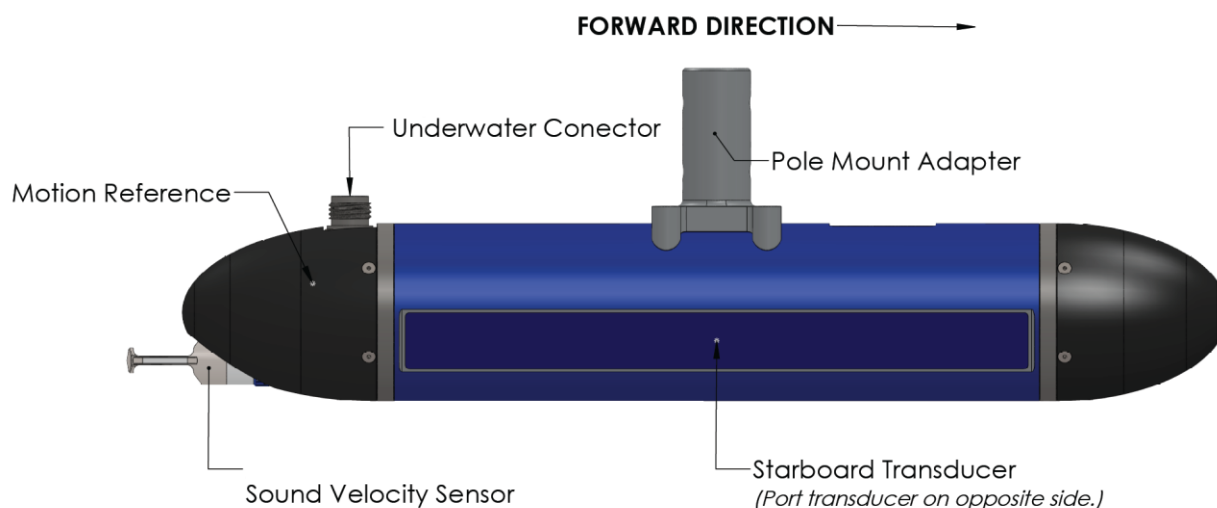
THIS DOCUMENT DESCRIBES HOW TO SET UP THE 3DSS-IDX SONAR AND SHOULD BE READ FULLY.

2	Contents.....	2
3	Software Installation and PC Requirements	3
4	Network Setup	3
5	Sonar Interface Unit Setup	4
5.1	SIU Power.....	4
5.2	SIU User Interface	5
5.3	SIU Serial Ports.....	5
5.4	SIU PPS and Trigger	6
5.5	LED Indicators	6
5.6	Integrated GNSS Antenna Connections.....	8
5.7	Ethernet Connections.....	8
5.8	3DSS Connection	8
6	Mounting the 3DSS-IDX Sonar Head	9
7	Configuring an External GNSS	10
8	Measuring the IMU to GNSS Antenna Lever Arms.....	10
9	3DSS-DX Software.....	11
10	Programming Ellipse INS to GNSS Antenna Offsets	18
10.1	Using the SIU user interface – Ellipse IMU	18
10.2	USING THE 3DSS CONTROL CONSOLE	19
11	Programming Ekinox INS to GNSS Antenna Offsets	20
12	Third Party Acquisition Software	23
13	How to Care for Your Sonar and Sonar Cable	24
14	Start-up Checklist.....	25
15	Troubleshooting	26
16	3DSS-IDX Sensor Positions And Offsets	28
17	Document History	29

2 CONTENTS

The following are included with the 3DSS-IDX:

- 3DSS-IDX-450 – Sonar Head with
 - AML MicroX Sound Velocity Probe SN: _____
 - SBG Ellipse-3 INS (if not Ekinox) SN: _____
 - SBG Ekinox- (if not Ellipse) SN: _____
- 3DSS-SIU – Sonar Interface Unit with
 - Septentrio AsteRx GNSS Receiver SN: _____
 - SBG Navsight INS Processor SN: _____
- 3DSS-DSC-10M – Sonar Cable
- 3DSS-SIU-PC150 – SIU DC Power Cable and alligator clips.
- 3DSS-SIU-PA90 – SIU AC/DC Power Adapter 90W
- 3DSS-SIU-FWC – SIU Firmware Update USB-RS232 Cable.
- GNSS L1/L2/L5 Antennas (2) and TNC cables (2).
- 3DSS-USB-KIT – Two 3DSS license dongles and USB flash with software.
- 3DSS-OR-KIT – O-ring kit for sonar cable.
- 3DSS-IDX-ZINC – Zinc anodes (2).



3 SOFTWARE INSTALLATION AND PC REQUIREMENTS

Locate the USB flash drive provided with your 3DSS-IDX sonar and prepare your acquisition computer by installing the following software found on the drive:

- 3DSS-DX Control: Controls the 3DSS DX or IDX sonar and is the data server for other applications.
- 3DSS-DX Display: Displays 2D and 3D sidescan data
- 3DSS-DX Profile: Displays the bathymetry cross-section profile¹
- 3DSS-IDX Binary Logger: Logs SBG INS and Septentrio GNSS raw binaries for post processing.
- 3DSS-OpenCPN Display: Chart display with real-time vessel location and swath area.

To run the Control and Display applications a license is required. The provided 3DSS dongles act as portable licenses. One dongle has both a Control and a Display license and two identical dongles are provided with the system.

To successfully run the above software suite, a Windows PC is required with the specifications below:

Minimum PC Specifications:

Intel i5 3rd Generation CPU
8 GB RAM
256GB HD
Gigabit Ethernet Adaptor
Intel Iris XE Integrated Graphics Processor

Recommend PC Specifications:

Intel i5 10th Generation Quad Core CPU or better
16 GB RAM
512GB SSD or larger
Gigabit Ethernet Adaptor
Discrete Graphics Adapter (NVIDIA GTX 1650 or better))

4 NETWORK SETUP

The 3DSS-DX/IDX sonar head uses a fixed IP address that cannot be changed.

Sonar Head IP Address:	192.168.228.1	fixed
------------------------	---------------	-------

The IP setup for the PC network adapter connected to the 3DSS Sonar Interface Unit (SIU) should have the following IP setup:

PC Static IP Address:	192.168.228.x	where x is any between 2 and 89
Netmask:	255.255.255.0	
Gateway:	empty	
DNS:	empty	
*Interface Metric:	5	

* Found under Ethernet Properties -> IPv4 Properties -> Advanced

The SIU, the optional integrated Septentrio GNSS receiver and the optional SBG Navsight INS Processor also provide Ethernet connectivity and are preconfigured with the below static IP addresses. User interfaces for these devices can be accessed through a web browser.

¹ On some Windows 10 machines you must run 3DSS Profile with graphics enabled.

SIU IP Address:	192.168.228.90
Septentrio IP Address:	192.168.228.91
SBG Navsight IP Address:	192.168.228.92

After the SIU has been powered on for approximately one minute, the sonar head will have completed its boot process and can be pinged using the Windows network ping utility (i.e. from a Windows Command Prompt).

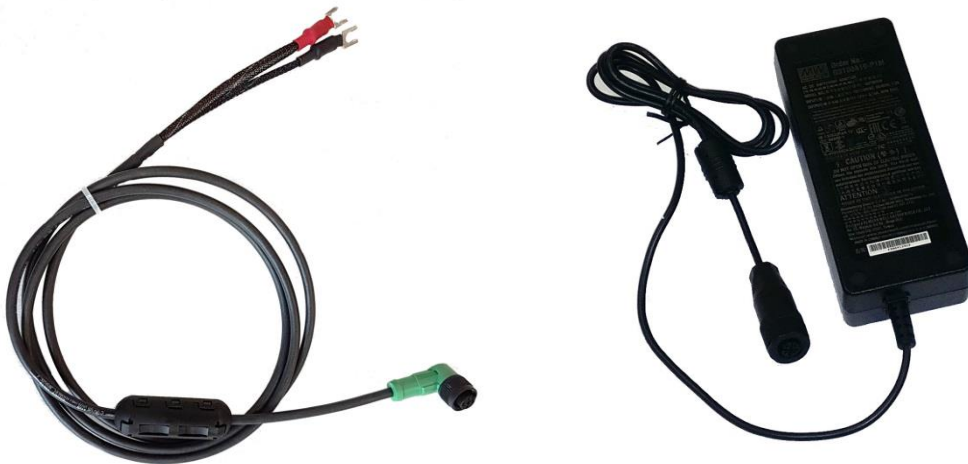
5 SONAR INTERFACE UNIT SETUP

The Sonar Interface Unit (SIU) is the blue aluminum box provided with the 3DSS-IDX sonar. The SIU powers the sonar head, has an integrated Ethernet switch for communication with the sonar head, and optionally has an integrated Septentrio GNSS receiver and SBG Navsight INS Processor.



5.1 SIU POWER

The SIU can be powered from a DC power source using the provided breakout cable, or from an AC power source using the provided AC/DC adapter (GST-90A).



To power the SIU from a DC source, the SIU must be provided with:

DC Voltage Range:	11.0V to 28.0 V
DC Voltage Nominal:	12V or 24V
DC Amperage Nominal:	8A @ 12V or 4A @ 24V

The SIU can be powered from a 12V battery or 12V DC regulated source, however care must be taken to keep the length of wires short (<1.5m) or gauge of wires sufficient to keep the resistance low. The provided DC breakout cable can be connected directly to 12V or 24V battery terminals.

5.2 SIU USER INTERFACE

The SIU user interface can be accessed through a web browser at <http://192.168.228.90>. This provides:

- Access to status information relating to the 3DSS Sonar, the SBG Ellipse or optional Ekinox IMU, and the optional Septentrio GNSS receiver.
- Configuration options for the SIU I/O ports.
- Logging of SBG Ellipse data for post processing².
User interface for configuring SBG Ellipse GNSS lever arms.

5.3 SIU SERIAL PORTS

If the optional integrated GNSS receiver is not included in the SIU, or if a user wishes to use an external GNSS, the first DB-9 port on the SIU labelled **GPS IN** can be used to provide the sonar with external NMEA-0183 GPS data at a baud-rate of 38,400 bps. Please refer to the full 3DSS Sonar Manual for details regarding the port interface wiring and the GPS NMEA sentence recommendations³.

By default, the internal SBG IMU (Ellipse or optional Ekinox) will provide motion data to the 3DSS sonar. However the second DB-9 port labelled **MRU IN** can optionally be used to provide the sonar with external TSS1 format motion data at a baud-rate of 115,200 bps. For further details please refer to the full 3DSS Sonar Manual.

The third DB-9 port labelled **AUX** has three different uses. The first is for SIU firmware upgrades using the provided USB-RS-232 cable and null modem cable / adapter (in the 3DSS-IDX Accessories Kit). Please refer to the 3DSS Sonar Manual for instructions regarding firmware updates.

The second use is for real-time input of GNSS RTK corrections. The AUX port internally connects to the Septentrio COM2 port which is preconfigured to detect RTCM v2 or v3 corrections at 115,200



² For the optional Ekinox IMU, IMU data logging is provided through the SBG Navsight web server instead of the SIU web server.

³ An external GNSS can be augmented with a PPS input for sonar time synchronization with GNSS time using the SIU PPS connector. Please refer to the 3DSS Sonar Manual for details.

baud. Using the provided USB-RS-232 cable RTCMv2 or v3 corrections can be easily input from a PC based NTRIP client with minimal setup required. Additionally, for NTRIP servers requiring precise position, a NMEA output can be configured on COM2 of the Septentrio. To change the Septentrio COM2 port configuration for alternative RTK correction inputs Septentrio provides a web interface that can be accessed at <http://192.168.228.91/>. Please refer to the 3DSS Sonar Manual for further details.

5.4 SIU PPS AND TRIGGER

Two 5V logic level connections are provided: PPS and TRIG. These are configurable as either input or output through the SIU User Interface.

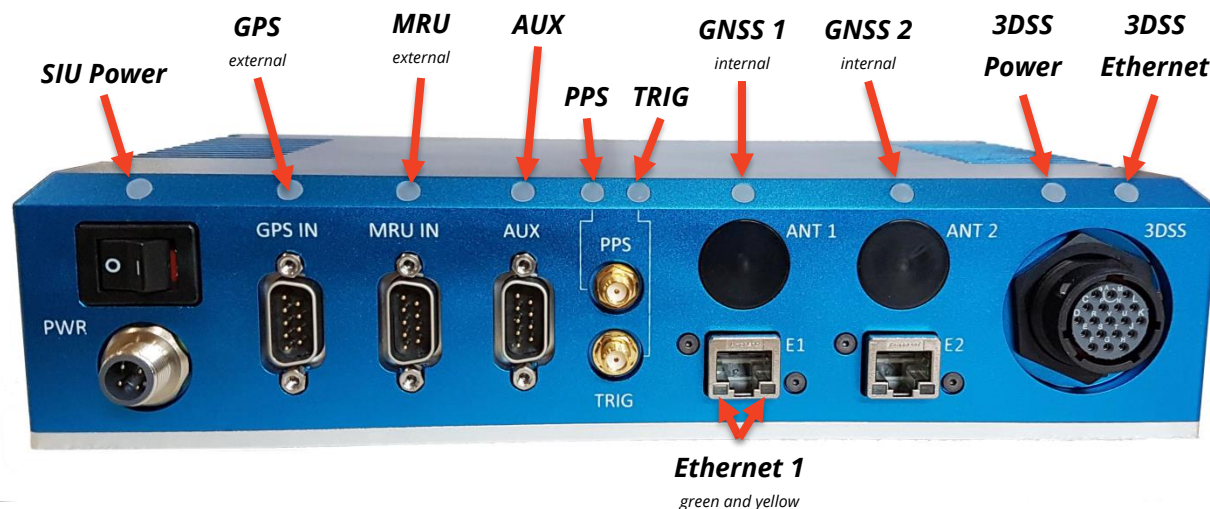
The PPS SMA connector when configured as an input (default) allows for the connection of an external 2.5V to 5.0V positive edge logic level pulse per second synchronization (PPS) signal. If an integrated Septentrio GNSS is installed, this PPS port can alternatively be configured as an output to share the internal Septentrio PPS signal with other devices.

The TRIG SMA connector when configured as an output (default) provides a 5.0V trigger output which transitions from low to high when the sonar starts its transmit cycle, and transitions from high to low when the receive cycle completes. Alternatively, if configured as an input an external 2.5V to 5.0V trigger source can be connected which will cause the sonar to transmit on the low to high transition of a pulse. Note: In this scenario the “Trigger” setting in the 3DSS-DX Sonar Control software must be set to “Software”.

5.5 LED INDICATORS

When the SIU is powered up, the 3DSS Power LED will be yellow while the sonar head is booting. The sonar head will not respond to ethernet communications from software during this time. It should also be noted that the SIU should not be switched on and off and on in rapid succession as this can cause a loss of all internal settings. Once the 3DSS Power LED becomes green the sonar head is ready for operation. The SIU LED is also yellow during a firmware update as detailed in the 3DSS Sonar Manual.

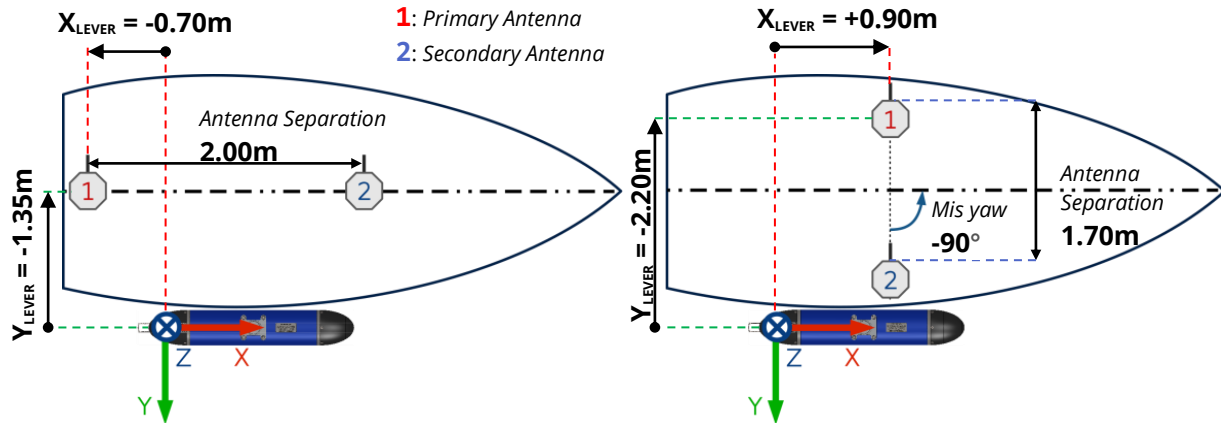
The remaining SIU LED's are summarized below and are mostly self explanatory. Note that the AUX LED only indicates the presence of incoming data, it does not indicate whether that data is valid for RTK corrections. Valid RTK data is indicated with the GNSS 2 LED. Note that although the GNSS 1 and GNSS 2 LED's are located above the GNSS Antenna connectors they indicate GNSS and RTK data activity respectively and do not verify the actual Antenna signals.



LED	STATE	MEANING
SIU Power	Off Yellow Green	SIU off – no power SIU in bootloader / firmware update SIU powered and operational
GPS (external) or GNSS 1 (internal)	Off Red Yellow Green	No data detected Invalid input data or no GNSS fix Incomplete GPS data GPS fix and is outputting all required data
MRU (external)	Off Red Green	No external MRU detected Invalid input data MRU TSS1 or NMEA detected
AUX	Off Yellow	No data detected Data detected
PPS	Off Yellow Green	No PPS detected Unstable PPS Stable PPS
TRIG	Off Green / blinking	Sonar is not pinging Sonar is pinging
GNSS 2 (internal)	Off Yellow Green	No RTK detected Float RTK Fixed RTK
Ethernet 1 & 2	Off Green Yellow	No Ethernet link Ethernet link up Ethernet data active
3DSS Power	Red Yellow Green	Power to sonar outside nominal range Sonar booting Sonar powered and operational
3DSS Ethernet	Off Green Green (flashing)	No Ethernet link Ethernet link up Ethernet data active

5.6 INTEGRATED GNSS ANTENNA CONNECTIONS

There are two antenna connections for IDX systems with the integrated Septentrio GNSS receiver. The ANT 1 connection is the primary antenna connection and ANT 2 is the secondary antenna. In the leftmost configuration below ANT 1 is placed aft of ANT 2 and their baseline is parallel to the direction of travel of the vessel with a 1m to 2m separation between them:



If preferred, the antennas can also be oriented in an athwartships port-starboard configuration and either side antenna can be the primary antenna. The ANT 1 connection on the SIU designates the primary antenna. The two GNSS antennas provided with systems that include the integrated Septentrio GNSS use 5/8-11UNC threaded mounting posts. Once mounted and connected to the SIU, the antenna locations relative to the 3DSS IMU location must be measured and entered into the system setup software. Please refer the **Measuring IMU to Antenna Lever Arms** section below for measurement details.

Note: For accurate positioning, the TNC antenna cables used to connect the two antennas to the SIU must be of the same type and length. The two TNC RF cables provided with the 3DSS-IDX with integrated GNSS are matched.

IMPORTANT: The primary antenna is always the one connected to ANT 1.

5.7 ETHERNET CONNECTIONS

The two Ethernet connections E1 and E2 are identical and connect to an Ethernet switch inside the SIU. The sonar acquisition computer should be connected to one of these two ports. A second computer (e.g. to run third party software) can be optionally connected to the system without the need to for a separate ethernet switch.

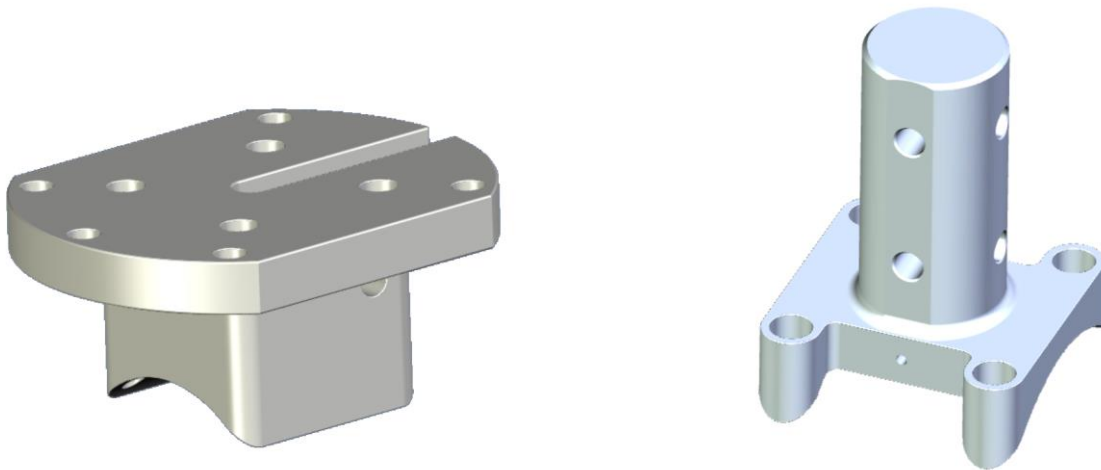
5.8 3DSS CONNECTION

The 3DSS-IDX sonar head must be connected to the 3DSS connector on the SIU using the provided cable.

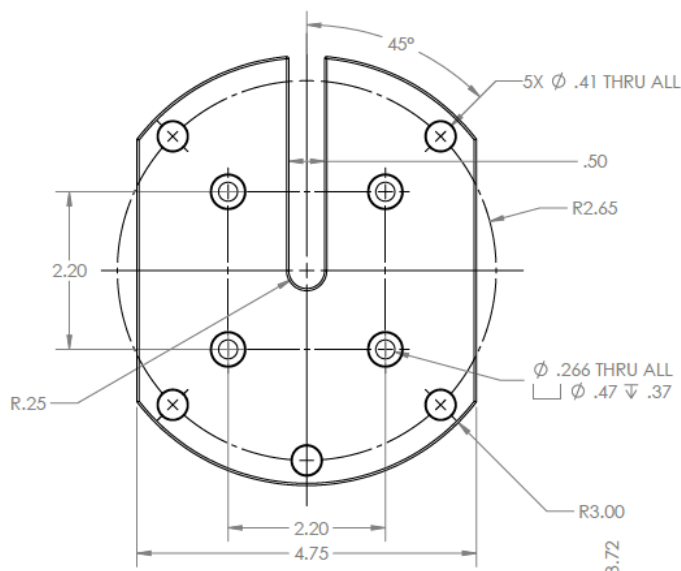
6 MOUNTING THE 3DSS-IDX SONAR HEAD

The 3DSS sonar head is designed to mount either over the port or starboard side of the vessel, or over the bow. The sonar head should be mounted 0.5m to 1.0m below the water line. The minimum mounting depth is determined by the draft of the vessel (e.g. to avoid multipath or acoustic shadowing from the hull when mounted over the side) and by motion or wave action that may expose the transducers to bubbles or air during operation since this will severely degrade results. The sonar head can be mounted deeper than 1.0m provided the mounting pole is large and does not wobble while the vessel is moving at the desired survey speed. The 3DSS-IDX sonar is designed to operate at survey speeds of 2 – 8 kt. The sonar head should also be mounted forward of props by 2m, or if not possible, as far forward as is feasible to avoid acoustic noise from the propellers.

The 3DSS sonar head can be mounted using either the default flange mount adapter, shown below on the left, or the included optional stem mount adapter, on the right.



The hole pattern for the default flange mount is shown below. For the hole pattern for the stem mount adapter please see the 3DSS Sonar Manual.



IMPORTANT: It is critical for the 3DSS sonar head to be mounted with a sufficiently sturdy pole (ideally with a fairing) to prevent pole wobble at the desired survey speed.

7 CONFIGURING AN EXTERNAL GNSS

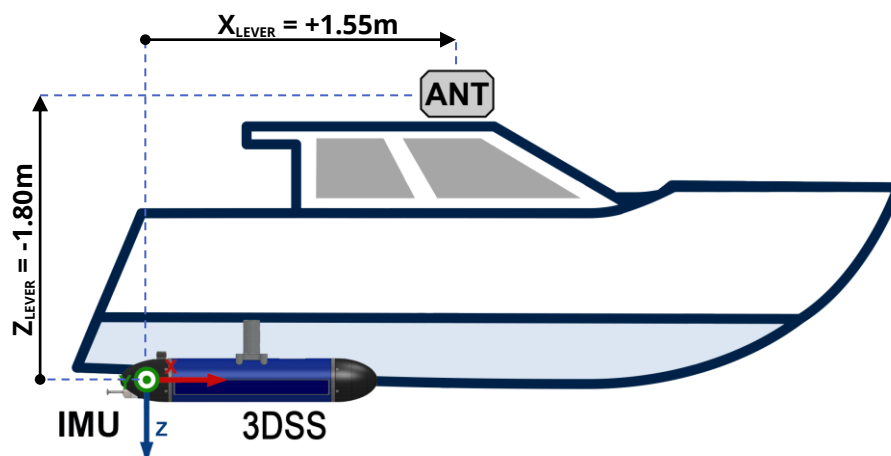
For proper operation with an external GNSS – the external GNSS must be configured and connected to the GPS port of the SIU. The GNSS must have two receivers (i.e. two antennas) to provide accurate heading and the GNSS must be configured to output the following NMEA-0183 sentences at a baud rate of 38,400 baud:

- \$GPRMC @ 5 Hz (or higher)
- \$GPGLL @ 5 Hz (or higher)
- \$GPGST @ 5 Hz (or higher)
- \$GPHDT @ 5 Hz (or higher) (if applicable, on dual antenna systems), and
- \$GPZDA @ 1 Hz.

NOTE: The internal Septentrio GNSS is pre-configured at the factory to output the NMEA sentences listed above but also outputs supplementary binary data over Ethernet UDP/IP. The binary data can typically be accepted directly into most third party survey software packages during acquisition as an independent navigation data stream (please see the **Third Party Software** section below) and can also be logged by 3DSS software for navigation post processing using SBG's Qintia PRO navigation processing software (please see the **3DSS Software** section below). If the SIU includes the optional integrated Septentrio GNSS, connecting an external GNSS will disable the internal Septentrio GNSS. If an external GNSS is used instead of the optional integrated Septentrio GNSS, logging of binary GNSS data is up to the User.

8 MEASURING THE IMU TO GNSS ANTENNA LEVER ARMS

The GNSS primary lever arm is the signed distance, expressed in the vehicle coordinate frame, **from** the internal IMU (Ellipse-3 / Ekinox-3) **to** the phase centre of the primary GNSS antenna,. The GNSS secondary lever arm is the signed distance, expressed in the vehicle coordinate frame, **from** the internal IMU (Ellipse-3 / Ekinox-3) **to** the phase centre of the secondary GNSS antenna. For best performance neither lever arm should exceed 10 m and both should be measured to within 5cm. Lever arms **must** be programmed into the system INS. Please see the section **Programming the GNSS Lever Arms** below.



Dual antenna GNSS systems, such as the integrated Septentrio GNSS, require the following in order to achieve optimal performance:

- Maximum separation between the two antennas should not exceed 3 m.
- The same antenna type and same cables with identical lengths must be used for both antennas.
- Both antennas must be mounted in the same orientation with respect to the vessel.
- Both antennas must have the same view of the sky when mounted on the vessel.

IMPORTANT: The 3DSS-IDX INS uses the antenna locations to determine the true heading value for both the port-starboard and the fore-aft antenna configurations and so there is no need to enter a heading offset into third party software when the SBG Navigation Driver is used for acquisition. Please refer to the section **Third Party Software** for details.

9 3DSS-DX SOFTWARE

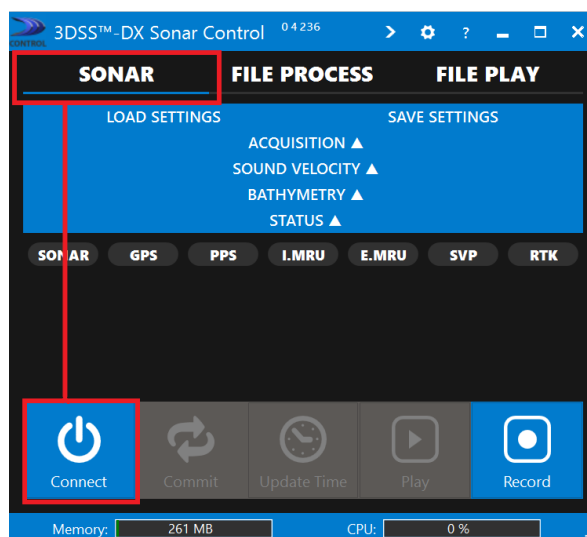
The 3DSS-DX Control software is used to operate the 3DSS sonar, process raw sonar data, and distribute the processed sonar data to other applications such as the 3DSS Displays and third-party survey packages.

- Plug the 3DSS USB Dongle into an available USB port on the acquisition computer
- Turn on the SIU and wait several seconds for the **3DSS Power LED** to turn green. Avoid turning the SIU off while this LED is yellow.
- Start the 3DSS-DX Control software application.

On the Sonar **Control** GUI do the following:

1. Select the **Sonar** Tab
(It will have a 'blue' underline when it is selected)

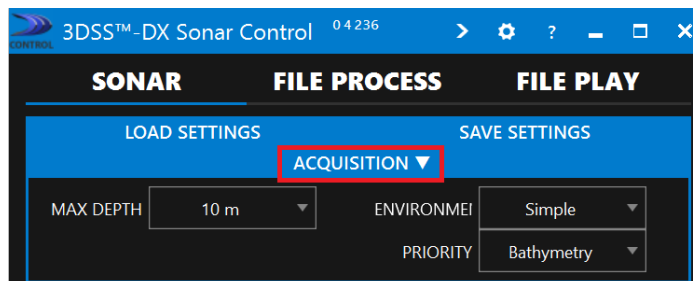
Click the **Connect** button to connect to the 3DSS sonar (power to the sonar head should have been established at least 60 seconds prior). If the connection to the Sonar head is successful the Commit icon beside **Connect** will turn blue.



ACQUISITION

2. Click on the **Acquisition** tab.
 - a. Set the **Depth** to an estimate of the maximum Depth setting for the survey.

NOTE: The 3DSS Sonar is NOT Auto-Ranging. However if the actual depth exceeds the **Depth** setting, data will not be clipped or lost. The swath width will be reduced however.



The recommended approach is to set the Depth to the MAXIMUM Depth necessary to accommodate the deepest water areas with the caveat that this also sets the maximum ping rate (i.e. data density). In cases where higher data density is desired in shallower areas, the Depth setting should be reduced appropriately.

- b. Set **Environment** to **Simple** or **Complex**

A **Simple** environment has an generally flat or slowly varying seabed and no significant vertical or water-column structures, while a **Complex** environment has a more rugged/variable seabed and/or includes vertical or water-column features (e.g. walls).

- c. Set **Priority** to the desired setting (Bathymetry, Wide Area Imaging, High Res Imaging).

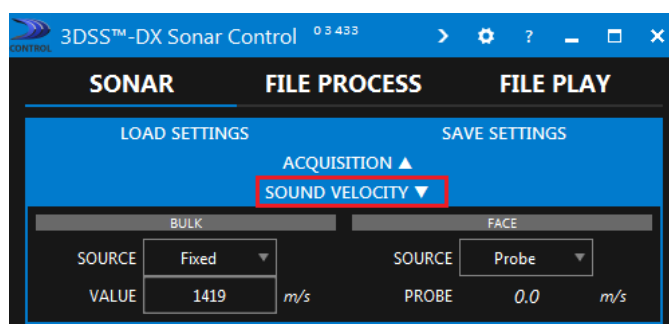
The priority setting optimizes settings for the specified priority but valid bathymetry and 3D/2D imaging data in all cases. The **Bathymetry** setting is optimal for hydrographic surveys while **Wide Area Imaging** is optimal for a search or rapid assessment application where a large area is to be covered as quickly as possible to find potential targets of interest. The **High Res Imaging** setting is optimal for getting the best possible image of a region of interest or an object with maximum detail.

SOUND VELOCITY

3. Click on the **Sound Velocity** tab and set the following.

BULK

- a. **Source:** Fixed



Value: Enter an estimate for the average sound velocity in the water-column. **Note:** The Bulk SV is used to calculate the range of the received data and is typically replaced by a measured sound velocity profile (SVP) in third party survey software. One or more SVP's are typically measured during a survey and imported into the 3rd party survey software. The value for the 3DSS Bulk SV is used to calculate range for the real-time sonar data (bathymetry and imagery).

FACE

b. **Face:** Probe

The real-time SV value measured by the integrated SV probe will be displayed below this setting when the sonar is pinging. This is a single point SV measurement at the sonar head and is used to compute the arrival angles of the received data. It is not altered in 3rd party survey software and real-time accuracy is therefore important.

Note: If 3DSS sonar data is recorded within the **Control** application, the real-time SV value can be modified after recording by using **Control's Reprocessing** Mode (please see the 3DSS Sonar Manual for more information about reprocessing).

IMPORTANT: The integrated SV probe has a small open cell that it uses to measure sound velocity and that must be kept free of bubbles and debris in order to maintain accuracy.

BATHYMETRY

4. Click on the **Bathymetry** tab
Select **MBES Mode** under the main dropdown, and set the following starting settings:

POINT REJECTION

c. **Min Depth:** 1m

The minimum depth rejects any soundings less than this depth.

d. **Swath:** 0x

The maximum swath width to accept as a multiple of depth. A setting of zero disables the swath width filter.

e. **Amplitude:** 0.3

The threshold factor for rejecting soundings based on their associated backscatter amplitude. The amplitude threshold is set by multiplying the specified value by the mean amplitude of the soundings for each ping. For example, a value of 1 means that all soundings that are below the mean of the amplitudes of the soundings for the same ping, are rejected.

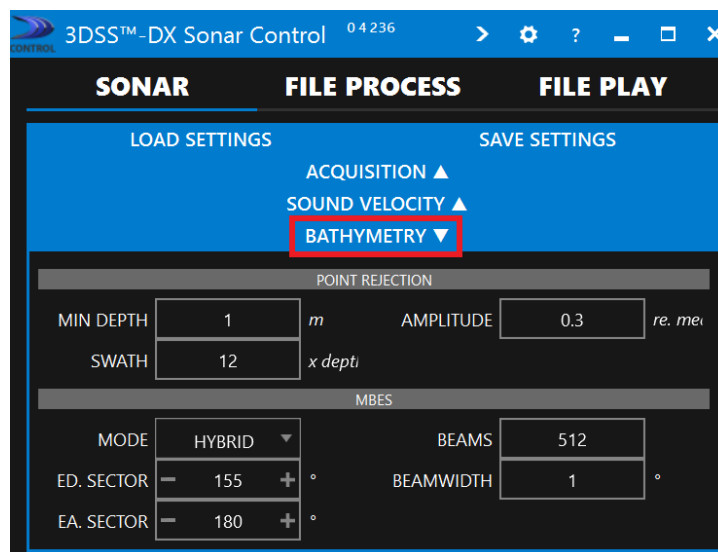
MBES

f. **Mode:** Hybrid

Specifies how the virtual beams will be distributed across the swath for a perfectly flat seabed. Hybrid is a combination of an Equidistant spacing and Equi-angle spacing.

g. **ED. Sector:** 155°

Specifies the angular extent of Equidistant beams.



h. **EA. Sector:** 180°

Specifies the full angular sector for all beams with the region beyond the ED Sector covered with beams distributed by Equi-angle.

NOTE: For a 180° sector either a Hybrid distribution or an Equi-angle distribution is required.

i. **Beams:** 512

Specifies the number of beams to use in the specified full angular sector.

j. **Beamwidth:** 1°

Specifies the beamwidth to use for the soundings.

NOTE: The beams are virtual but are also ideal in that they have no sidelobes and are all identical, unlike MBES beams. Please refer to 3DSS Sonar Manual for more information about Bathymetry settings.

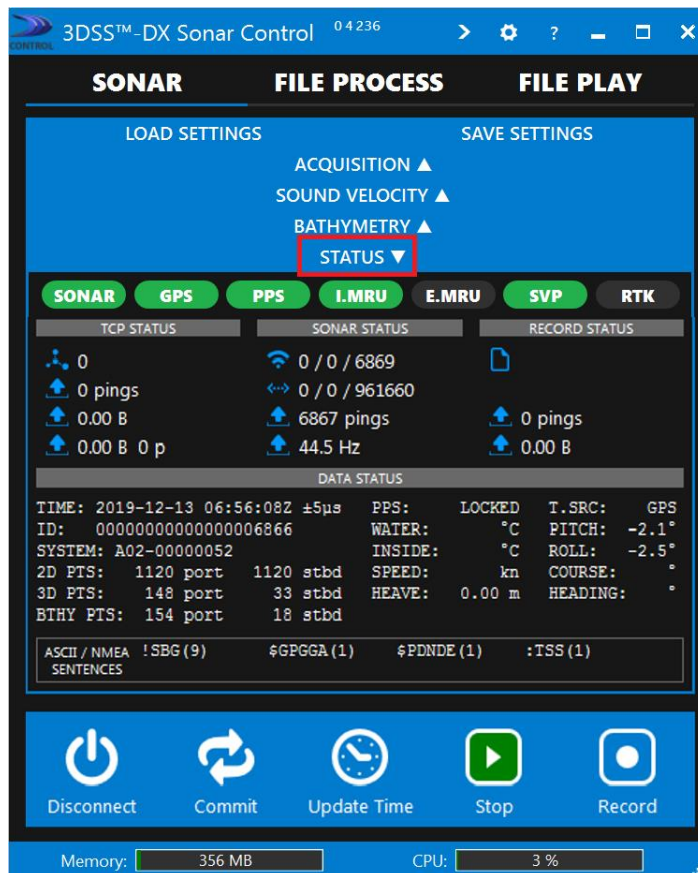
STATUS

5. Click on the **Status** tab.

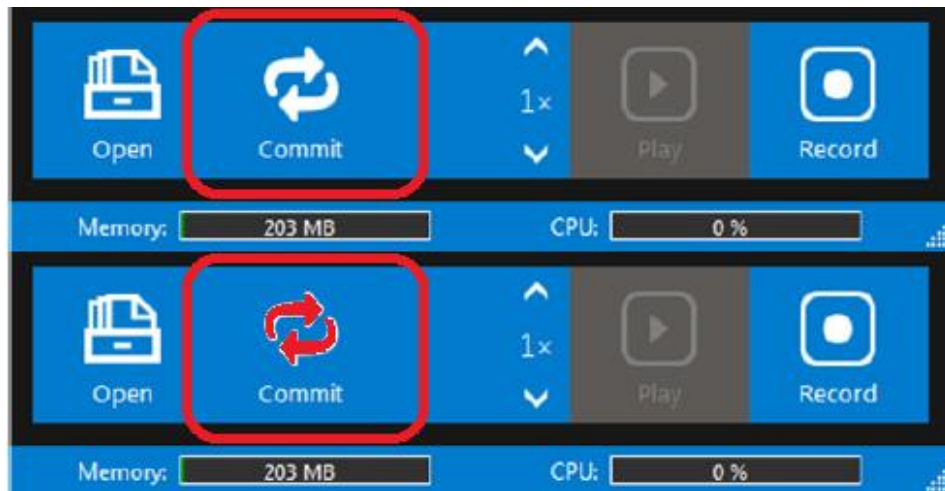
Verify that **ALL** Device Status Indicators are **GREEN**. If a Device is **YELLOW** or **RED**, the data from that device has an issue or is not getting to the sonar. Hovering over the device indicator will provide a message indicating the issue.

NOTE: A device indicator is black if the device is not in use. For example, the E.MRU status indicator for the E.MRU is black if an external MRU is not connected which is typical for 3DSS-IDX systems)

NOTE: The I.MRU indicator will be yellow initially until the SBG INS is 'aligned' which means it has not yet reached acceptable accuracy. INS alignment requires motion (e.g. figure eight vessel paths at approximately 4 knots). Alignment will typically require less than 10 minutes of motion.

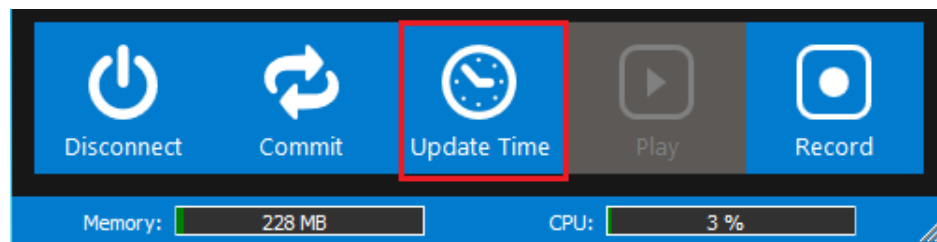


6. Click on **Commit** to send the above settings to the sonar.

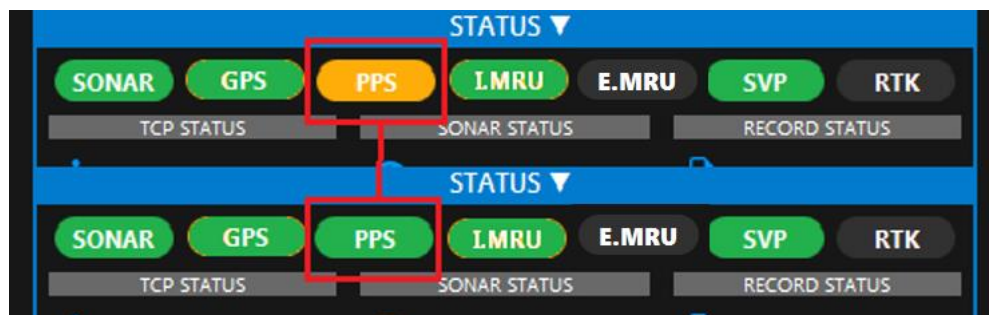


(IMPORTANT NOTE: Each time a 'setting/parameter' is modified in any of the above Tabs, the '**Commit**' button will begin blinking **RED** / **WHITE** and MUST be clicked, in order to send the modified setting/parameter to the sonar.)

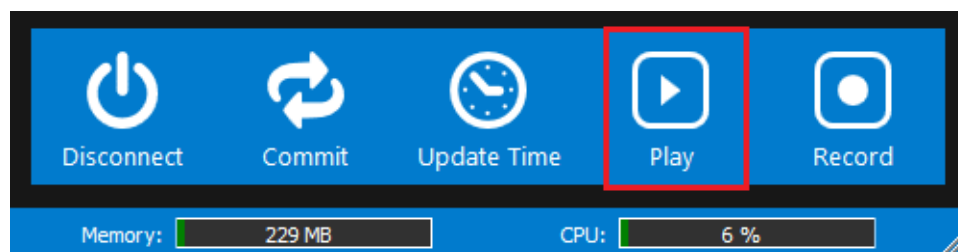
7. Click **Update Time** to send the current time to the sonar.



(IMPORTANT NOTE: If the PPS light is **YELLOW** and not **GREEN**, then, you may need to click the '**Update Time**' button again, as the timing from the GPS / INS may have NOT been fully initialized, when it was first clicked, and the sonar did NOT get the correct timestamp.)



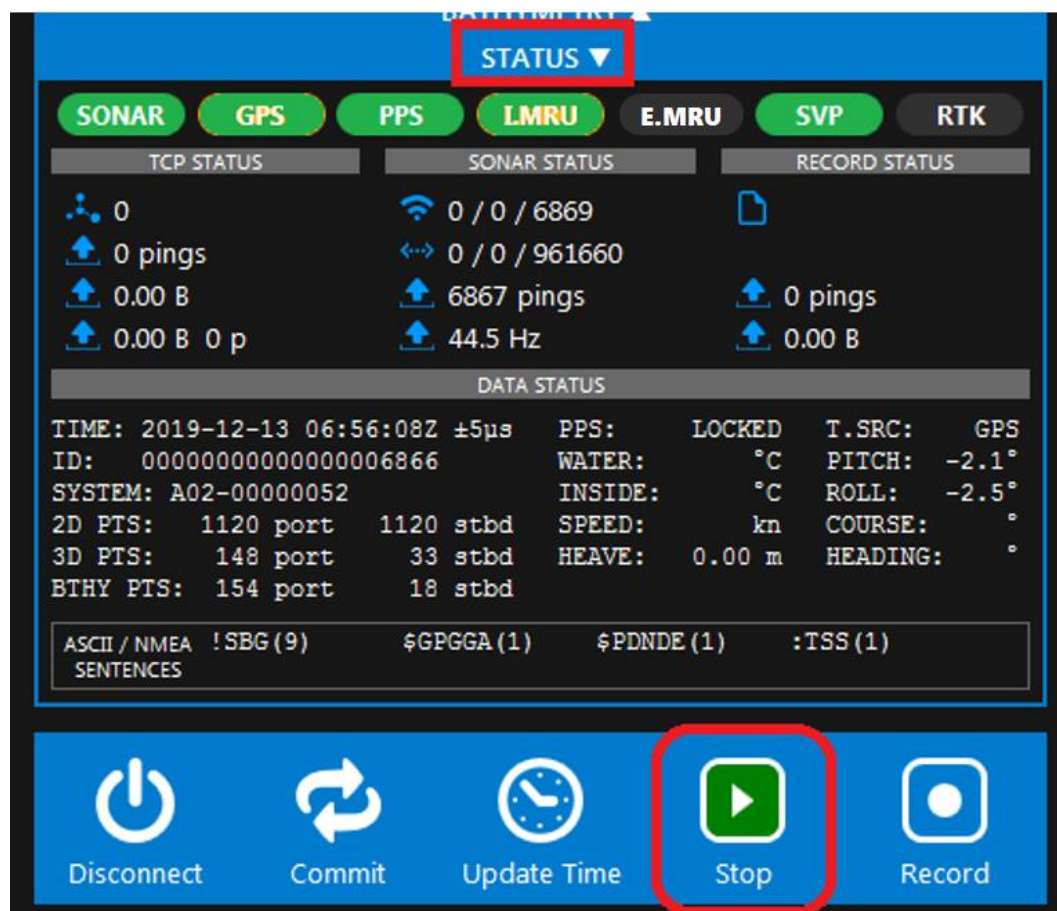
8. Click **Play** to start pinging.



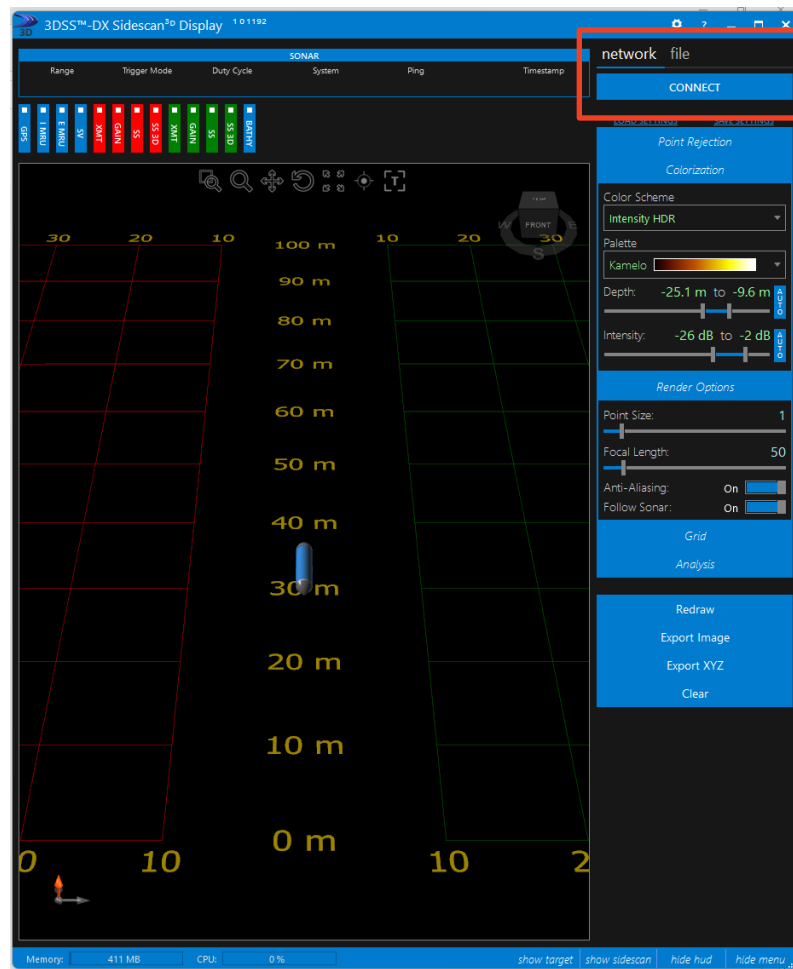
The sonar is now pinging and you should see information from each ping being displayed under the **Status** tab.

The **RED / YELLOW / GREEN** lights just below **Status** are indicators and hovering the mouse cursor over top can get more information and diagnose setup issues.

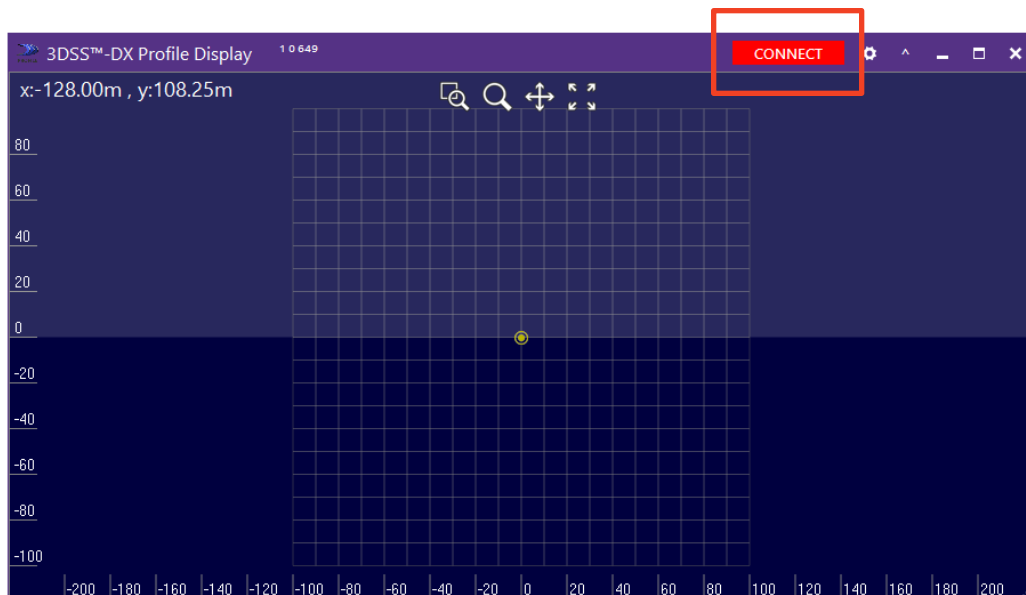
If you need to stop Pinging, click on the 'Stop' button



9. Start 3DSS-DX Display and click **Network** and then **Connect** in the top right corner of the application. You should now be able to see the 3D sonar data point cloud.



10. Start 3DSS-DX Profile and click **Connect** on the top right. You should now be able to see the sonar bathymetry data displayed as cross-sections.



10 PROGRAMMING ELLIPSE INS TO GNSS ANTENNA OFFSETS

This section is specific to the 3DSS-IDX with the Ellipse-3 INS. Once the lever arms are measured they must be programmed into the Ellipse-3 through either the SIU user interface, or through the 3DSS Control Console interface. The lever arms is specified in **meters** and **degrees**.

The lever arm values are stored inside the Ellipse-3 and needs to be updated when the installed orientation of the sonar head or GPS antennas changes thereby changing the lever arms.

10.1 USING THE SIU USER INTERFACE – ELLIPSE IMU

1. With the sonar head plugged into the SIU, power on the SIU and load the SIU user interface through a web browser at <http://192.168.228.90>
2. Navigate to the “Ellipse” tab. A dialog box should pop up indicating that the Ellipse settings were successfully read and values should be populated for the primary and secondary lever arms. If not, try clicking “Refresh” at the bottom of the page.
3. Enter the measured lever arms to the primary and secondary antennas. Secondary antenna lever arms are by default entered as (X,Y,Z) coordinates but can alternatively be entered as (pitch, yaw, antenna separation) by selecting the associated checkbox.
4. Select the “GNSS Aiding Source” as either “External NMEA” if GNSS aiding is coming from the external GPS port on the SIU, or “Internal Septentrio” if GNSS aiding is coming from the integrated Septentrio GNSS receiver.
5. Click “Save” at the bottom of the page. A dialog box should appear indicating the settings were saved successfully.



Primary Antenna Lever Arm

X Offset (m)	Y Offset (m)	Z Offset (m)
0.00	0.00	0.00

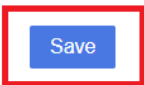
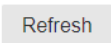
Secondary Antenna Lever Arm

☒ Enter as (x, y, z) ☐ Enter as (pitch, yaw, separation) ☐ Single Antenna Only

X Offset (m)	Y Offset (m)	Z Offset (m)
1.000	0.000	0.000

Aiding Options

GNSS Aiding Source: Internal Septentrio  **4**

 **5** 

10.2 USING THE 3DSS CONTROL CONSOLE

1. Turn on the sonar head and connect to it using 3DSS Control.
2. Click the chevron in the top-right of the 3DSS Control application to open up the command console.



3. You will now be able to enter command in the console to query the Ellipse-3 and also check and set the lever arm. See marker 3.
4. Run **sbg --get-offsets** to check the offsets programmed into the Ellipse-3.
5. Run **sbg --set-offsets=x:y:z:pitch:yaw:antenna** to update the lever arm values stored in the Ellipse-3. Where **x** is the *Xlever* offset, **y** is the *Ylever* offset, **z** is the *Zlever* offset, all specified in meters, and pitch is the **pitch** misalignment in degrees (typically 0), **yaw** is the yaw misalignment in degrees, and **antenna** is the antenna separation distance specified in meters.

For example, to set the lever arm of *Xlever* = +0.90m, *Ylever* = -2.20m, and *Zlever* = -1.80m in a single antenna system run:

```
sbg --set-offsets=0.90:-2.20:-1.80
```

For a dual-antenna system you will have the additional parameters of yaw misalignment and antenna separation distance. For a yaw misalignment of -90° and an antenna separation of 1.70m, run the following command:

```
sbg --set-offsets=0.90:-2.20:-1.80:0:-90:1.70
```

6. Finally, you **must save the settings** in the Ellipse-3 by running the following command:
sbg --save
7. After a couple seconds – you can run the **sbg --get-offsets** command again to confirm that the correct offsets were written to the Ellipse-3.

Figure 1 on the next page shows the 3DSS Control application after all of the commands in steps 4 – 7 have been executed.

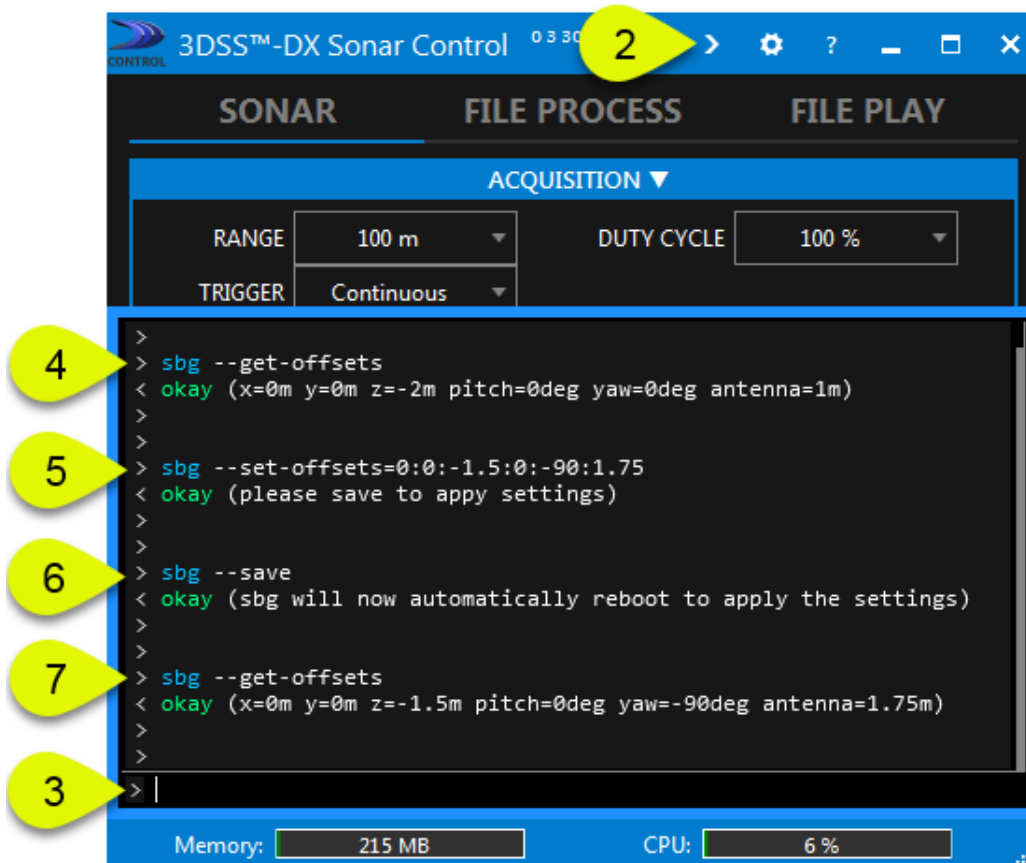


FIGURE 1. PROGRAMMING ELLIPSE-3 LEVER ARM USING COMMAND CONSOLE

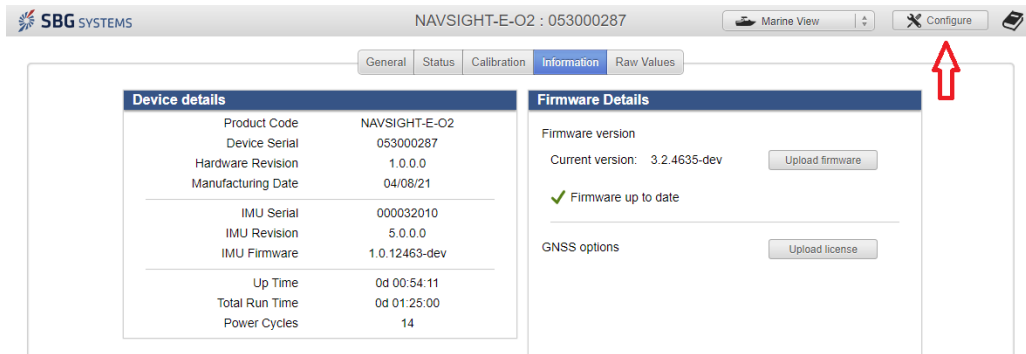
11 PROGRAMMING EKINOX INS TO GNSS ANTENNA OFFSETS

This section is specific to the 3DSS-IDX with the Ekinox-3 INS. Once the lever arm is measured it must be programmed into the Ekinox-3 through the use of SBG's Navsight web interface. The web interface can be accessed at: <http://192.168.228.92/>

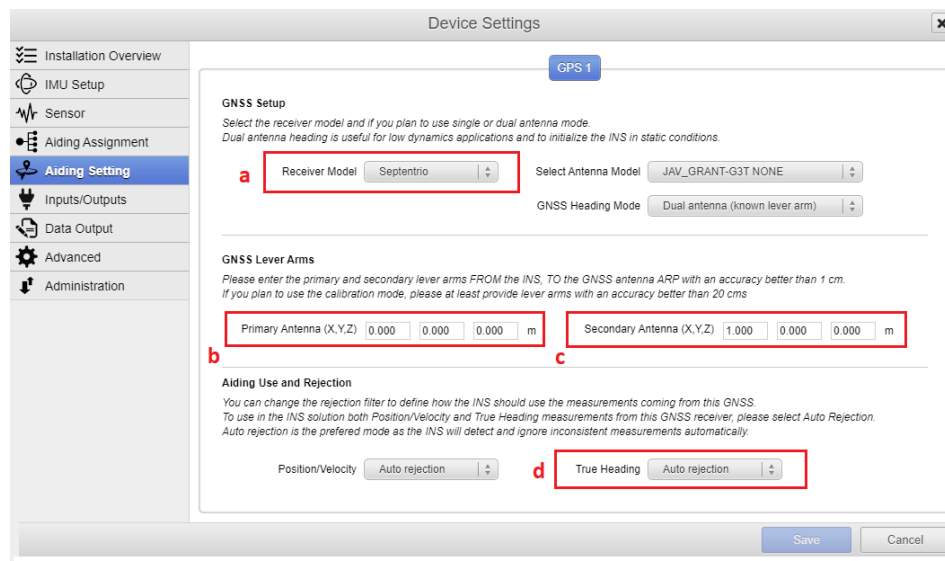
The lever arm is specified in **meters** and **degrees**. The lever arm value is stored inside the Navsight Ekinox-3 INS and needs to be updated when the installed orientation of the sonar head or GPS antennas changes thereby changing the lever arm.

To program the lever arm into the Ellipse-3 inside the iDX follow these steps:

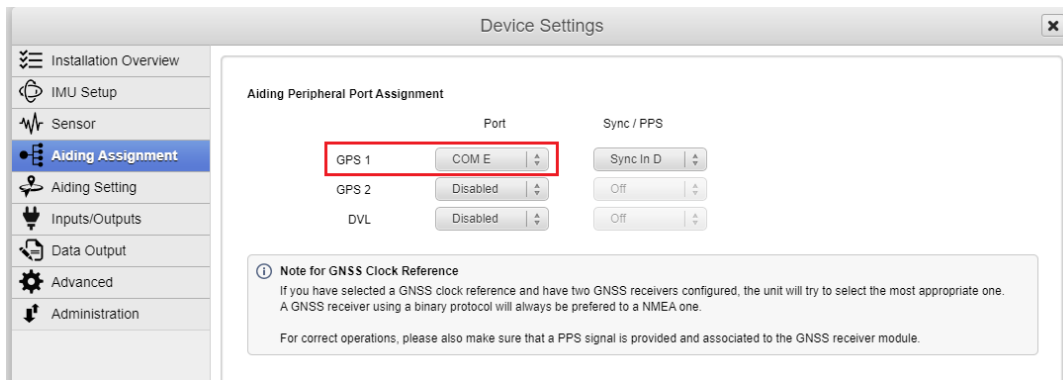
1. Turn on the 3DSS sonar interface unit with the sonar head connected.
2. Use a web browser to open the Navsight web interface at <http://192.168.228.92/>
3. Click on Configure in the top right corner of the Navsight web interface.



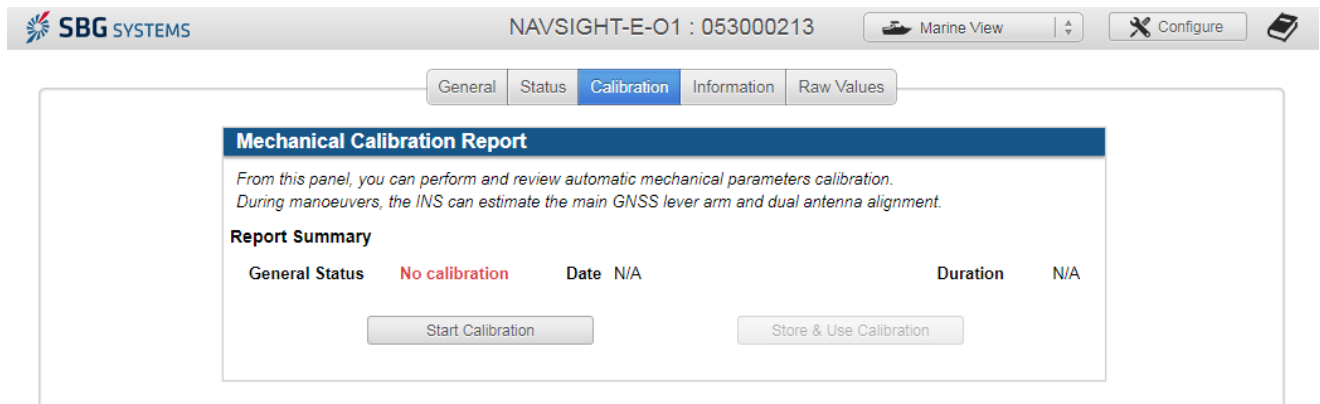
- In the new window, click the Aiding Settings tab on the left and configure GPS 1 with the lever arm for the primary and secondary antennas with the lever-arms measured in Section 8.



- The GPS model should be set to Septentrio if the 3DSS SIU has an internally integrated Septentrio Asterx-4 GNSS. Otherwise, the model should be set to NMEA. This setting is correctly programmed at the factory and should not require a change.
 - The Primary Antenna lever arm must be programmed as measured in Section 8.
 - The Secondary Antenna lever arm must be programmed as measured in Section 8.
 - The True Heading should be set to Auto rejection if the SIU has an integrated Septentrio Asterx-4 GNSS. Otherwise, if an external GNSS is being used for aiding Auto rejection should be configured as Always accept.
- Under the Aiding Assignment tab configure the GPS 1 port as COM E if an integrated Septentrio Asterx-4 is installed in the SIU and is being used for aiding, otherwise the GPS 1 port should be configured as COM D for external aiding.



6. As a final step – you may run the Navsight calibration. The calibration takes anywhere from 20 minutes (with RTK) to 60 minutes (without RTK). Calibration must be performed with the system moving and will validate the lever arms and antenna alignment.



12 THIRD PARTY ACQUISITION SOFTWARE

Sonar, position, and motion data can be interfaced into compatible third-party software for real-time acquisition. Depending on the sonar model and configuration, different data sources may be available. Generally, data will either be sourced from the 3DSS-DX Control software, from the sonar head itself, or from an integrated GNSS / INS sensor if applicable. Note that the reference location may be different depending on which data source is used for position and motion information.

The 3DSS-DX Control software makes available a TCP stream containing 3D sidescan, bathymetry, 2D sidescan, as well as GNSS position data (NMEA-0183), and motion data (TSS). In the third-party acquisition software this can be utilized and configured through a 3DSS or PingDSP driver with the below settings.

IP Address: 127.0.0.1* Port: 23848 Protocol: TCP

* If the acquisition software and 3DSS-DX Control software are not running on the same PC, the IP address above should instead be set to the IP address of the PC running 3DSS-DX Control.

The position and motion information in the above data stream is a timestamped copy of the GNSS (NMEA-0183) and MRU (TSS) data from either the external (if an external GNSS or MRU is present) or the integrated GNSS and motion sensor. In the case of integrated sensors, the position will be referenced to the primary GNSS antenna, and the TSS output will be referenced to the center of the sonar. Note that this position is the position from the GNSS, not the EKF solution from the INS.

Users with a 3DSS-IDX will likely want to take advantage of the INS solution that is available from the integrated SBG sensor. In this case, it is recommended to not use the position or motion data mentioned above, and instead use the SBG's INS solution directly. In the third-party acquisition software this can be done by adding and configuring an SBG Ekinox device. Configure the IP address and ports as below.

For an IDX-Base or IDX-FULL:

IP Address: 192.168.228.1 Port: 24333 Protocol: UDP

For an IDX-PRO:

IP Address: 192.168.228.92 Port: 24333 Protocol: UDP

The SBG INS can be used for both position (lat, long, altitude), motion (pitch, roll, heave), as well as heading and will be referenced to the INS EKF position shown in the diagram in Section 16 of this manual.

13 HOW TO CARE FOR YOUR SONAR AND SONAR CABLE

The 3DSS-DX is ruggedly designed and built, and therefore requires little maintenance. However, to ensure long-lasting, reliable service, some periodic maintenance is recommended.

The 3DSS-DX / IDX is not designed for long-term deployments, and it is not recommended to keep the system in the water for more than two weeks at a time. If the system is to be deployed for 2+ days of continuous submersion – the provided zinc anode should be installed. After each deployment, the system should be rinsed thoroughly with freshwater, and its port and starboard transducer arrays inspected for any organism growth or damage. To remove any buildup of debris, the transducer faces should be washed using a soft cloth and a mild non-abrasive hand soap with water.

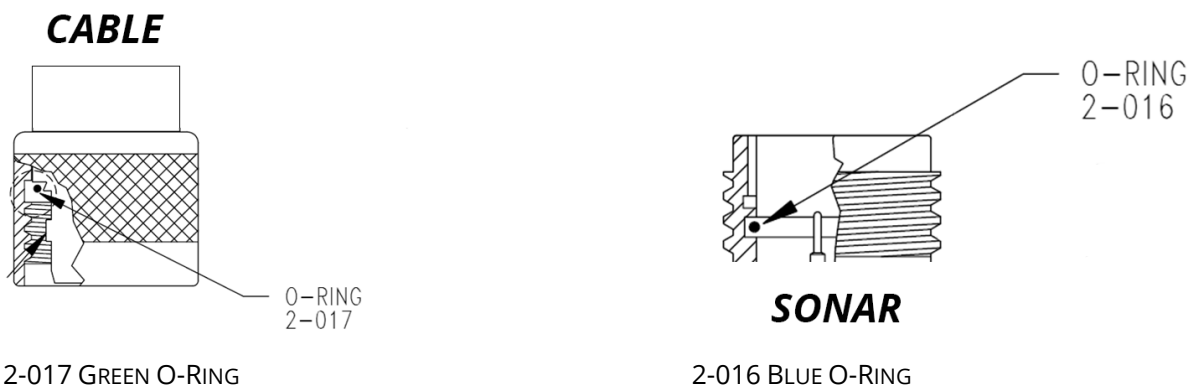
IMPORTANT: DO NOT USE HARSE CHEMICAL DETERGENTS OR SOLVENTS ON THE TRANSDUCER FACES.

Any time the sonar cable is disconnected from the sonar head the O-rings and connector pins in both the cable end and connector end should be inspected for any grime buildup. The connector on the sonar cable should be clean and inspected for the presence of a single green 2-017 O-ring. The connector on the sonar head should be clean and inspected for the presence of a blue 2-016 O-ring. If grime buildup does occur, the O-rings should be removed the connectors cleaned with isopropyl and a low lint Q-tip, and new greased O-rings must be installed. To extend the life and increase the reliability of the O-ring seal, apply a thin film of an O-ring lubricant before installation of the O-ring. The underwater connector on the sonar unit, as well as, the mating end on the underwater cable should always be kept clean and protected when the cable is not connected. The underwater connector must be dried before disconnecting the underwater connector from the sonar unit to prevent any salt water ingress into the connector. If salt water does come in contact with the inside of the connector then use fresh water, isopropyl, and a low lint Q-tip to clean the connector before mating it again.

UV EXPOSURE WARNING: Do not allow the 3DSS sonar head to sit in direct sunlight when not in use. Prolonged exposure to UV will damage the polyurethane face of transducers.

In order to protect the hardware components of the sonar the following should be observed:

- DO NOT connect or disconnect the main sonar cable with the power on.
- DO NOT attempt to open and service the 3DSS sonar head.
- DO NOT attempt to open and service the Sonar Interface Unit unless specifically instructed to by Ping DSP.
- DO NOT place the 3DSS sonar head where the transducers may be scratched or damaged.
- DO NOT exceed the bend radius the main sonar cable (100mm).
- DO NOT clean the sonar head using a pressure washer – the high-velocity spray will cut through the polyurethane transducer face.



14 START-UP CHECKLIST

The following is a check list that should be executed each time the 3DSS-IDX is mobilized on a new platform:

1. Check that the underwater connector is clean, and has both O-rings installed.

There should be a green O-ring in the underwater connector on the sonar cable end, and a blue O-ring in the underwater connector on the sonar head.
2. Check that the top-side connector is clean.
3. After powering the SIU:
 - (a) Confirm that the Ethernet link between the SIU and sonar head is active.
The right-most 3DSS Ethernet LED on the SIU should be solid green or flashing green.
 - (b) Confirm that the Ethernet link between the SIU and acquisition computer is active.
The green LED on port E1 or E2 should be lit.
4. Confirm the correct INS to GNSS primary antenna lever arms have been applied in the SBG INS.
5. Confirm the primary antenna is connected to ANT 1, and the secondary antenna connected to ANT 2.
6. On the acquisition PC firewalls should be disabled, and the IP address of the network adapter used to communicate with the sonar should be configured to be on the 192.168.228.XX subnet. If multiple network adapters are used on the PC, the Interface Metric of the network adapter for the sonar should be set to '5', and other network adapter IP settings should not overlap with the 192.168.228 subnet.

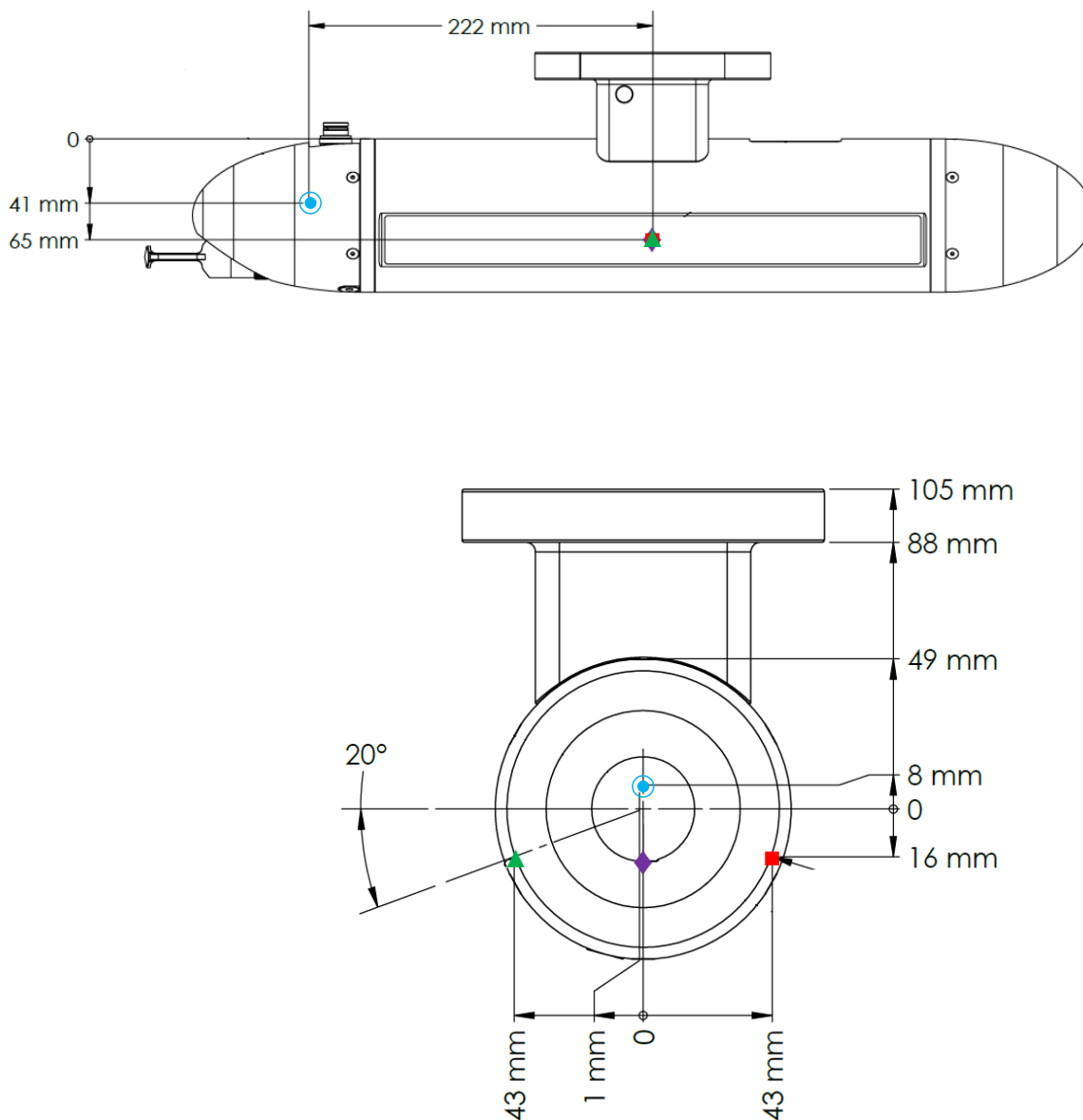
15 TROUBLESHOOTING

<i>Symptom</i>	<i>Potential Causes</i>	<i>Corrective Action</i>
Power LED on the SIU is off.	The power switch is not turned on.	Verify the power switch is in the on position.
	No power is applied.	Verify the SIU is connected to a suitable power source. If so, check power source.
	Supply voltage is too low or too high.	Check that the DC supply voltage is between 11.5V and 28.5V.

<i>Symptom</i>	<i>Potential Causes</i>	<i>Corrective Action</i>
Cannot Connect to the Sonar	The sonar is not being supplied power, or the sonar is not connected to the SIU.	Verify the left sonar LED on the SIU is green. Power-cycle the SIU if it is off or red. Verify that the sonar connection between the SIU and the 3DSS sonar is fully made. Make sure connectors at both ends of the sonar cable are free of debris and water.
	Network is not setup correctly.	Verify the acquisition computer is connected to one of the Ethernet ports on the SIU using a Cat-5E Ethernet cable. Verify Ethernet link is active by viewing a green LED on SIU E1 or E2 port, and that the right sonar LED is lit. Verify the Ethernet adapter on the acquisition computer is enabled and has a static IP configuration. The IP address should be set to 192.168.228.XXX (where XXX is not 001), and the subnet mask set to 255.255.255.0.
	Internal software or Firmware crash.	Turn off power to the sonar and restart the 3DSS Control Software. Next, power the sonar, wait 30 seconds and then try to connect through the 3DSS Control software.
	Firewall Blocking the Sonar	Disable the Windows Firewall and/or other firewalls on the acquisition PC associated with the LAN adapter used by the sonar. If this fixes the issue either leave the firewall(s) disabled, or add an exception to the firewall for the 3DSS-IDX Control Application.
	Multiple Ethernet Adapters Interfering with Sonar Operation.	Change the advanced Interface Metric network adapter setting from Auto to 5 for the ethernet adapter connected to the 3DSS topside SIU.

<i>Symptom</i>	<i>Potential Causes</i>	<i>Corrective Action</i>
GPS, PPS, or MRU data is not displayed in the Control Software Status.	Sonar is not connected and/or not pinging.	Connect to the 3DSS sonar through the Control Application. Then Commit -> Update Time -> Play to start pinging. (NMEA settings come in through ping data).
	Device is not connected properly, or not powered.	Ensure the external MRU, GPS, or PPS device is powered and connected to the SIU via an RS232 DB9 connector. If it is, the signal could be coming in on pin 2 of the DB9 connector; try using a suitable RS232 crossover cable. Check connection between SIU and sonar and make sure both connectors are free of debris and water.
	Baud Rates Incorrect	(Applies to MRU and GPS only) The GPS device should be set to a baud rate of 38,400 and the MRU devices should be set to 115,200 baud. Contact technical support if this is not possible.

16 3DSS-IDX SENSOR POSITIONS AND OFFSETS



- The position of the center of measurement for the SBG INS.
- The acoustic center of the port transducer.
- ▲ The acoustic center of the starboard transducer.
- ◆ The reference for INS EKF processed positions output by the SBG INS.

17 DOCUMENT HISTORY

Revision	Date	Author	Brief Description
1.0	2018-11-14	MPB	First release.
1.1	2019-01-10	MPB	Corrected instructions for setting SBG-GPS lever arm. Added external GPS configuration instructions. Added info for RTCM / CMR corrections input.
1.2	2019-04-02	MPB	Added cover page. Added info about VBS.
1.3	2019-05-16	MPB	Added more information to Care and Maintenance. Fixed sonar drawing to remove connector type.
1.3a	2020-03-03	MPB	Fixed small pagination error.
1.4	2021-04-20	MPB/KN	Added more text to care for sonar section. Added SBG Navsight information and lever arm configuration. Added SIU web interface information, input / output operation of PPS and TRIG.
1.5	2021-06-07	MPB	Updated guide to match IDX-BASE/FULL upgrade to Ellipse-V3.
1.6	2022-08-15	KN	Added section for general third-party acquisition setup Added yellow LED case for the 3DSS Power LED
1.7	2024-04-18	MR	Updated guide to reflect new simple GUI

Please contact support@pingdsp.com if any questions arise.

Ping DSP Support
support@pingdsp.com

